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Thermodynamic description of the Mg–Eu binary system

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ABSTRACT

Thermodynamic assessment of the Mg–Eu binary system has been carried out by combining first-principles calculations and Miedema's theory with CALPHAD method. Firstly, the mixing enthalpy of the liquid alloys was calculated by using Miedema's theory, and formation enthalpies of the intermetallic compounds were calculated by using the projector augmented-wave (PAW) method within the generalized gradient approximation (GGA). Subsequently, the liquid phase was described employing a simple substitutional model, of which the excess Gibbs energy was formulated with a Redlich-Kister expression. And the solubility of Eu in HCP_(Mg) and Mg in BCC_(Eu) were neglected. While the intermetallic compounds $Mg_{17}Eu_2$, Mg_5Eu, Mg_4Eu, Mg_2Eu and MgEu, were treated as stoichiometric compounds. Consequently, a set of self-consistent thermodynamic parameters for all stable phases in the Mg–Eu binary system were obtained, which can reproduce most of the thermodynamic and phase boundary data.

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1. Introduction

Mg-base alloys are being used in a diverse range of applications for their unique physical and mechanical properties, especially the low density, high specific strength and stiffness. Taking account of both specific Young's modulus and high specific strength, magnesium alloys show similar or even better values than Al-base alloys and many commercial steels [1]. In order to further improve on the mechanical properties of Mg-base alloys, various elements are often added into Mg-base alloys. Among these elements, Rare earths (RE) are of major significance enhancing the mechanical properties at elevated temperature and improving casting characteristics of Mg-based alloys [2]. In order to better understand the role of RE in the Mg-based alloys, thermodynamics and phase relations of the related systems are essential.

During the past decades, most of the Mg-RE binary systems [3–14] have been assessed by CALPHAD method. In this work, mixing enthalpy of liquid Mg–Eu alloys is calculated by using Miedema's theory, and formation enthalpies of the intermetallic compounds calculated through first-principles calculation based on the density function theory (DFT). Then thermodynamic description of the Mg–Eu system is assessed with the CALPHAD approach.

2. Evaluation of experiment information

The phase diagram of the Mg–Eu binary system was investigated by Mühlpfordt and Klemm [15], and four intermetallic compounds, $Mg_{17}Eu_2$, Mg_5Eu, Mg_2Eu and MgEu, along with the corresponding liquidus, three eutectic reactions and two peritectic reactions were determined. These data are all used in the present optimization.

The liquidus and solidus for the hcp (Mg) were also measured by Rokhlin [16]. And the maximum solubility of Eu in hcp Magnesium was determined to be about 4.8×10^{-5}at.% [16], which is too small thus neglected in the present work. As for the liquidus of hcp reported by Rokhlin [16], because of its unreasonable trends near to the eutectic points, they were not used in the present assessment.

Later, a new compound Mg_4Eu was detected by high resolution electron microscopy and X-ray powder diffraction [17]. However, whether this Mg_4Eu is formed congruently or incongruently remains unknown [17]. Although intergrowth of $Mg_{23}Eu_6$ in the Mg_4Eu matrix was also observed [17], $Mg_{23}Eu_6$ is still treated as Mg_4Eu in the present work because their compositions lie very close to each other and necessary information such as their melting temperature, crystal structure and reaction type etc. are lacking.

There were no thermodynamic parameters reported about Mg–Eu binary system. Hence, before assessment, first-principle and Miedema methods are respectively utilized to determine the formation enthalpies of the compounds and mixing enthalpy of liquid alloys.

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Table 1
Crystal structures and formation enthalpies of the Mg–Eu compounds by first-principle calculation

Phase	Pearson symbol	Space group	Prototype	Lattice (Å)		ΔH (kJ/mol)	Reference
				a	*c*		
Mg	hP2	$P6_3/mmc$	Mg	3.1946	5.1804	–	Present
				3.189	5.169	–	[29]
				3.2092	5.2105	–	[30]
$Mg_{17}Eu_2$	hP38	$P6_3/mmc$	Th_2Ni_{17}	10.4289	10.2629	−5.22	Present
				10.49	10.33	–	[31]
				10.468	10.374	–	[32]
				10.493	10.327	–	[33]
Mg_5Eu	hP42	$P6_3/mmc$	$EuMg_5$	10.3953	10.7463	–	[24]
	hP36	$P6_3/mmc$	$ErZn_5$	10.3597	10.6103	−4.23	Present
				10.4126	10.7626	–	[26]
Mg_4Eu	hP90	$P6_3/mmc$	Mg_4 Sr	10.416	28.051	–	[17]
Mg_2Eu	hP12	$P6_3/mmc$	$MgZn_2$	6.3319	10.2282	−12.16	Present
				6.379	10.308		[31]
				6.386	10.324		[34]
				6.3443	10.2585		[35]
				6.344	10.258		[36]
MgEu	cP2	$Pm\bar{3}m$	CsCl	4.0493	–	−9.20	Present
				4.102			[31]
Eu	cI2	$Im\bar{3}m$	W	4.4292	–	–	Present
				4.429			[29]
				4.5827			[37]

3. First-principles calculations and Miedema's method

The first-principles calculations were performed using the scalar relativistic all-electron Blöchl's projector augmented-wave (PAW) [18,19] method within the generalized gradient approximation (GGA), as implemented in the highly-efficient Vienna ab initio simulation package (VASP) [20,21]. For the GGA exchange-correlation function, the Perdew-Wang parameterization (PW91) [22,23] was employed. A plane-wave energy cutoff of 350 eV was held constant for four intermetallic compounds, $Mg_{17}Eu_2$, Mg_5Eu, Mg_2Eu and MgEu. Brillouin zone integrations were performed using Monkhorst-Pack K-point meshes and reciprocal space (k-point) meshes are increased to achieve convergence to a precision of 0.3 kJ/mol of atom. The k-point meshes for $Mg_{17}Eu_2$, $Mg_5Eu(ErZn_5$-type), Mg_2Eu and MgEu were $4\times4\times4$, $15\times15\times15$, $10\times10\times8$ and $15\times15\times15$, respectively. The total energy was converged numerically to less than 1×10^{-6} eV/unit with respect to electronic, ionic and unit cell degrees of freedom, the latter two relaxed using Hellman-Feymann forces with a preconditioned conjugated gradient algorithm. After structural optimization, the total forces on each ion were less than 0.01 eV/Å. All calculations are performed using the "high" setting within VASP to avoid wrap-around errors. Spin polarization was used in all calculations.

The formation enthalpies for the intermetallic compounds were calculated by the following equation

$$\Delta H\left(Mg_xEu_y\right) = E_{total}\left(Mg_xEu_y\right) - xE_{total}\,(Mg) - yE_{total}\,(Eu) \quad (1)$$

where $E_{total}\left(Mg_xEu_y\right)$, E_{total} (Mg), and E_{total} (Eu) are the calculated total energies (per atom at $T = 0$ K) of the four intermetallic compounds, pure Mg and pure Eu, respectively. The $EuMg_5$-type Mg_5Eu [24,25] could not be calculated because of the unfilled sites. The $ErZn_5$-type Mg_5Eu [26] was calculated in the present work, and the calculated formation enthalpy indicated that $ErZn_5$-type Mg_5Eu is not a stable phase, so the formation enthalpy has not been used in the optimization. For the Mg_4Eu phase, there are 90 atoms in the unit cell, which is too large to be calculated by our current computer cluster. Finally, in this work only the formation enthalpy of $Mg_{17}Eu_2$, Mg_2Eu and MgEu were calculated and used in the optimization.

Miedema's theory is an empirical thermodynamic theory [27], which can be applied to investigate the thermodynamic properties of binary alloys. According to the Miedema's theory, the mixing enthalpy of binary liquid alloys can be expressed as:

$$\Delta H = 2x_A f_B^A V_A^{2/3}\frac{-P(\Delta\phi^*)^2 + Q(\Delta n_{WS}^{1/3})^2 - R}{(n_{WS}^A)^{-\frac{1}{3}} + (n_{WS}^B)^{-\frac{1}{3}}} \quad (2)$$

where the f_B^A indicates the degree to which one type of atom (A) is surrounded by dissimilar neighboring atoms (B), $\Delta\phi^*$ is the chemical potential difference of the pure elements, Δn_{WS} is the electronic density difference of the boundary of Wigner–Seitz unit cell, n_{WS} is the electronic density of the boundary of pure element unit cell, and the P, Q and R are empirical parameters. The scheme of Miedema's theory has been described in detail [28,29].

4. Thermodynamic model

Using substitutional solution model, the Gibbs free energy of the liquid phase can be expressed as:

$$G^\phi = \sum {}^\circ G_i^\phi \cdot x_i + RT\sum x_i \ln x_i + {}^{ex}G^\phi \quad (3)$$

where G^ϕ is the Gibbs energy of the ϕ phase, ${}^\circ G_i^\phi$ the lattice stability of the element i, x_i the gross mole fraction of the element i in ϕ state, and ${}^{ex}G^\phi$ is the excess Gibbs energy of the solution phase, which is formulated by Redlich-Kister expression:

$${}^{ex}G^\phi = x_{Eu}x_{Mg}\sum_{j=0}^{n}\left(x_{Eu} - x_{Mg}\right)^j \cdot {}^{(j)}L_{Eu,Mg}^\phi \quad (4)$$

where ${}^{(j)}L_{Eu,Mg}^\phi$ is the interaction parameter between element Eu and Mg, calculated with ${}^{(j)}L_{Eu,Mg}^\phi = A_j + B_jT$, here A_j and B_j are to be optimized.

$Mg_{17}Eu_2$, Mg_5Eu, Mg_4Eu, Mg_2Eu and MgEu are all treated as stoichiometric compounds. Due to lack of the heat capacity data, Gibbs free energies for these phases are formulated with the assumption of Neumann-Koppe rule as following:

$$G^{Eu_aMg_b} = a\,{}^\circ G_{Eu}^{BCC} + b\,{}^\circ G_{Mg}^{HCP} + A + B\cdot T \quad (5)$$

where a and b are corresponding number in the formula for the compounds, and A and B are parameters to be optimized.

Table 2
Thermodynamic parameters of the Mg–Eu system

Phase	Thermodynamic parameters
Liquid	${}^{0}L_{\mathrm{Mg,Eu}} = -28\,188.0986 + 4.9471\,T$ ${}^{1}L_{\mathrm{Mg,Eu}} = 12\,085.2832 - 1.9185\,T$
$Mg_{17}Eu_2$	$G^{Mg_{17}Eu_2} = 0.8947^{\circ}G^{HCP}_{Mg} + 0.1053^{\circ}G^{BCC}_{Eu} - 5606.8445 + 1.0797\,T$
Mg_5Eu	$G^{Mg_5Eu} = 0.8333^{\circ}G^{HCP}_{Mg} + 0.1667^{\circ}G^{BCC}_{Eu} - 8597.4315 + 2.3833\,T$
Mg_4Eu	$G^{Mg_4Eu} = 0.8000^{\circ}G^{HCP}_{Mg} + 0.2000^{\circ}G^{BCC}_{Eu} - 9444.0000 + 2.2251\,T$
Mg_2Eu	$G^{Mg_2Eu} = 0.6667^{\circ}G^{HCP}_{Mg} + 0.3333^{\circ}G^{BCC}_{Eu} - 12\,682.5993 + 1.4106\,T$
MgEu	$G^{MgEu} = 0.5000^{\circ}G^{HCP}_{Mg} + 0.5000^{\circ}G^{BCC}_{Eu} - 9765.0520 + 1.2845\,T$

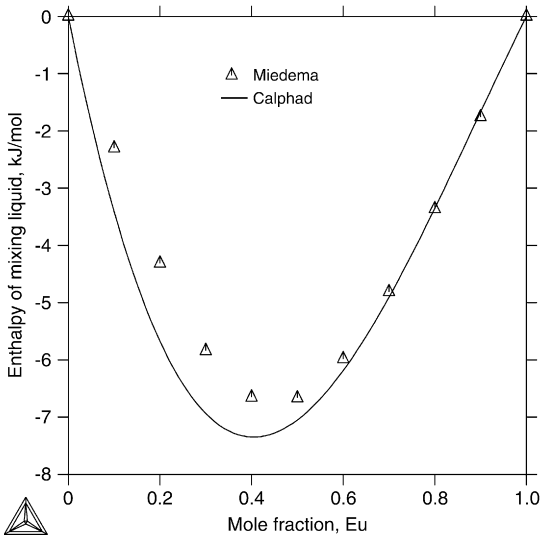

Fig. 1. Calculated mixing enthalpy of Mg–Eu liquid alloy at 1200 K compared with Miedema's data.

5. Results and discussion

Crystal structures, calculated lattice constants and formation enthalpies of the intermetallic compounds are listed in Table 1. The lattice constants of pure elements were previously calculated with first principles by Wang et al. [29]. As can be seen, the results obtained in the present calculation are in good agreement to Wang et al. [29]. The calculated lattice constants of the four intermetallic compounds are consistent with the experiment data from literature [30–37]. Considering the lattice constants of this system, it is reasonable to accept the formation enthalpy of intermetallic compounds obtained by first-principle calculation.

Using the lattice stabilities given by Dinsdale [38] and the Thermo-Calc software [39], the thermodynamic parameters (Table 2) of all the phases of Mg–Eu binary system were optimized on the basis of the calculated thermodynamic data through Miedema, first-principle calculations along with experimental phase equilibria. It can be seen later that good agreement has been realized between calculated and experimental values of this system.

Because few experimental thermodynamic data were reported about this system, the Miedema's theory was employed to calculate the mixing enthalpy of liquid Mg–Eu alloys. These calculated values were used as a starting value for optimizing. Comparison of the calculated mixing enthalpies by CALPHAD with the Miedema's theory at 1200 K is shown in Fig. 1. It can be seen that the lowest value of the assessed mixing enthalpy locates at the 40 at.% Eu, which is near the Miedema's location at about 45 at.% Eu. Considering the reasonable error of Miedema's method, the present assessment may be accepted.

Fig. 2 shows a comparison between the formation enthalpies of compounds obtained using first-principle calculations and

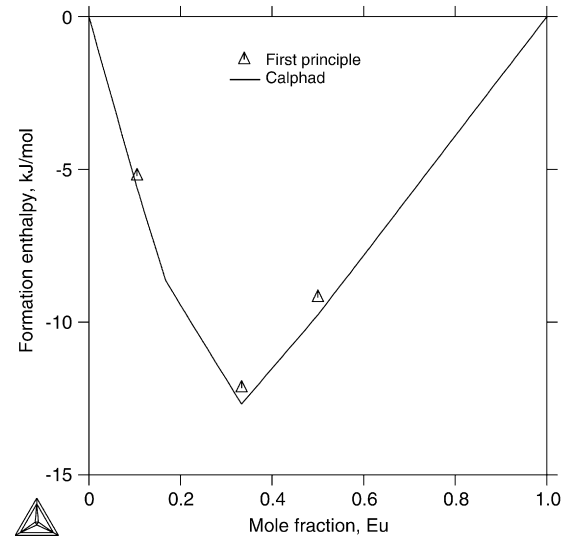

Fig. 2. Formation enthalpy of the Mg–Eu compounds calculated in this work.

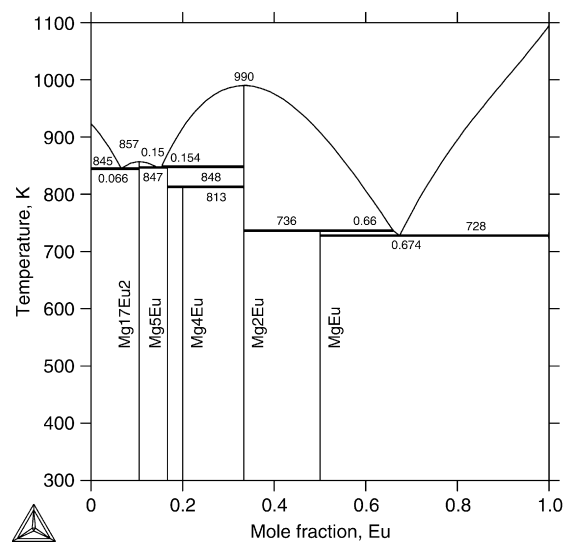

Fig. 3. The Mg–Eu phase diagram assessed in the present work.

CALPHAD assessment. Due to lack of corresponding experimental data of the compounds, high weight is set to the first-principle calculations during the optimization.

With the assessed parameters, phase diagram of the Mg–Eu system was further calculated as illustrated in Figs. 3 and 4. Together with Table 3 listing all invariant reactions, we see that the temperatures deviation between the calculated values and experimental data are within 10 °C for most invariant reactions except that involving Mg_4Eu. As mentioned before, the Mg_4Eu was only observed in samples annealed at 813 K for 6 weeks [17]. Although its stable range of temperature was not determined, we may know from the reference that it should exist at temperature lower than that of the reaction $L + Mg_2Eu \leftrightarrow Mg_5Eu$. However, due to lack of experiment, we assumed in present work that Mg_4Eu is formed by a eutectoid reaction at 813 K.

6. Conclusions

The Mg–Eu binary system has been thermodynamically assessed based on experimental data and first-principles calculations of the phase equilibrium and thermodynamic properties. A set of self-consistent thermodynamic parameters have been obtained.

Table 3
Summary of invariant reactions

Reaction	T (K)	Composition of liquid (at.% Eu)	Reaction type	Values in optimization	Reference
	844	0.6	Eutectic	Yes	[15]
$L \leftrightarrow HCP + Mg_{17}Eu_2$	835	0.7	Eutectic	No	[16]
	845	0.66	Eutectic	–	Present
$L \leftrightarrow Mg_{17}Eu_2$	864	10.53	Congruent	Yes	[15]
	857	10.53	Congruent	–	Present
$L \leftrightarrow Mg_{17}Eu_2 + Mg_5Eu$	842	14	Eutectic	Yes	[15]
	847	14.55	Eutectic	–	Present
$L + Mg_2Eu \leftrightarrow Mg_5Eu$	846	15.66	Peritectic	Yes	[15]
	848	15.42	Peritectic	–	Present
$Mg_5Eu + Mg_2Eu \leftrightarrow Mg_4Eu$	813	20	Unknown	Yes	[17]
	813	20	Eutectoid	–	Present
$L \leftrightarrow Mg_2Eu$	992	33.33	Congruent	Yes	[15]
	990	33.33	Congruent	–	Present
$L + Mg_2Eu \leftrightarrow MgEu$	736	66	Peritectic	Yes	[15]
	736	66	Peritectic	–	Present
$L \leftrightarrow MgEu + Eu$	722	69	Eutectic	Yes	[15]
	728	67.4	Eutectic	–	Present

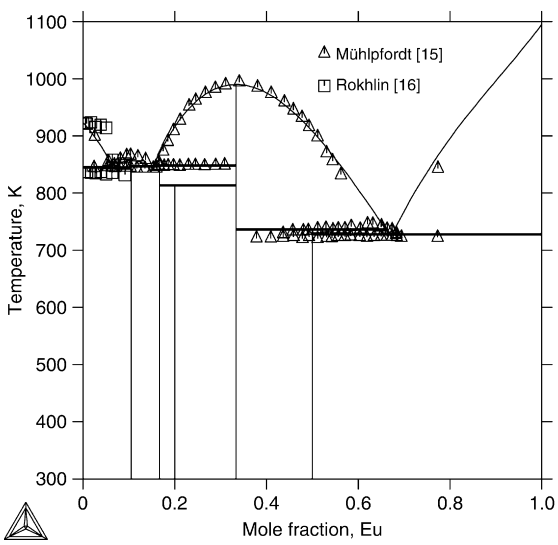

Fig. 4. Comparison between the measured phase diagram and the current calculation.

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